

1 **Supplementary material for “Fusing short-pulse voltage waveforms**
2 **with handcrafted features for rapid state-of-health estimation of**
3 **lithium iron phosphate batteries: a controlled comparison of encoder**
4 **pairings”**

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7 **Contents**

- 8 Supplementary Note S1. Handcrafted feature definitions
9 Supplementary Note S2. PatchTST encoder equations
10 Supplementary Note S3. Statistical test families
11 Supplementary Note S4. Re-implementation used for the generalisation, robustness
12 and deployment experiments
13 Supplementary Algorithm S1; Supplementary Tables S1–S6; Supplementary Figs.
14 S1–S5

15 **Supplementary Note S1. Handcrafted feature definitions**

16 This note defines the 143 waveform-derived features of Table 2; the seven
 17 context quantities are given in Eq. (3) of the main text. The segments follow the
 18 publisher's pulse timing and are defined, together with the first and second
 19 differences, as

$$20 \quad S^{(1)} = (V_1, \dots, V_{30}), S^{(2)} = (V_{31}, \dots, V_{40}), S^{(3)} = (V_{41}, \dots, V_{100}), d_t^{(1)} = V_{t+1} - V_t, d_t^{(2)} = d_{t+1}^{(1)} - d_t^{(1)} \\ 21 \quad \quad \quad (S1)$$

23 **Basic operators.** For a window x of length n ,

$$24 \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \sigma(x) = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}, R M S(x) = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}, E(x) = \sum_{i=1}^n x_i^2 \quad (S2)$$

$$25 \quad R(x) = \max x - \min x, I Q R(x) = Q_{0.75} - Q_{0.25}, \\ M A D(x) = \text{median}(|x - \text{median}(x)|), A U C(x) = \sum_{i=1}^{n-1} \frac{x_i + x_{i+1}}{2} \quad (S3)$$

26 The standard deviation uses the population form $(1/n)$, except for the global
 27 voltage sequence, which follows the publisher's supplementary definition with the
 28 sample form $(1/(n-1))$; both are labelled in the feature list.

29 **Shape.** Bias-corrected Fisher–Pearson skewness and excess kurtosis are used,
 30 with m_k the k -th central moment:

$$31 \quad m_k = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^k, g_1 = \frac{m_3}{m_2^{3/2}}, g_2 = \frac{m_4}{m_2^2} - 3, \\ G_1 = \frac{\sqrt{n(n-1)}}{n-2} g_1, G_2 = \frac{n-1}{(n-2)(n-3)} [(n+1)g_2 + 6] \quad (S4)$$

32 **Histogram entropy.** The range of the window is divided into 16 equal bins and
 33 p_i is the fraction of samples in bin i :

$$34 \quad H(x) = -\frac{1}{\ln 16} \sum_{p_i > 0} p_i \ln p_i \quad (S5)$$

35 **Linear trend.** A least-squares line is fitted against the sample index t_i :

$$a = \frac{\sum_i (x_i - \bar{x})(t_i - \bar{t})}{\sum_i (t_i - \bar{t})^2}, b = \bar{x} - a\bar{t}, R^2 = 1 - \frac{\sum_i (x_i - b - at_i)^2}{\sum_i (x_i - \bar{x})^2} \quad (S6)$$

Zero crossings and total variation. Zero crossings of a difference sequence are counted after removing exact zeros, giving y :

$$Z(y) = \#\{i: y_i y_{i+1} < 0\}, TV(x) = \sum_{i=1}^{n-1} |x_{i+1} - x_i| \quad (S7)$$

The 15 operators of Eqs. (S2)–(S7) applied to $S^{(1)}$, $S^{(2)}$ and $S^{(3)}$ give the segment statistics; applied to the full sequence and to $d^{(1)}$ and $d^{(2)}$, together with end points and quantiles, they give the 24 global and 22 difference-dynamics features.

Spectral features. The demeaned sequence is Hann-windowed and transformed by a real fast Fourier transform ($N=101$, frequency in cycles per sample):

$$w_t = 0.5 - 0.5 \cos\left(\frac{2\pi t}{N-1}\right), x_t = (V_t - \bar{V}) w_t, \quad (S8)$$

$$X_k = \sum_{t=0}^{N-1} x_t e^{-j2\pi kt/N}, P_k = |X_k|^2, f_k = \frac{k}{N}$$

The amplitudes of the first eight non-DC lines and the energy shares of four bands are

$$A_k = \frac{2|X_k|}{\sum_t w_t} (k=1, \dots, 8), \quad (S9)$$

$$B_1 = \frac{\sum_{k=1}^3 P_k}{\sum_{k=1}^{50} P_k}, B_2 = \frac{\sum_{k=4}^7 P_k}{\sum_{k=1}^{50} P_k}, B_3 = \frac{\sum_{k=8}^{15} P_k}{\sum_{k=1}^{50} P_k}, B_4 = \frac{\sum_{k=16}^{50} P_k}{\sum_{k=1}^{50} P_k}$$

Spectral-shape features exclude the DC component: dominant frequency and amplitude, centroid, spread, flatness, entropy and 85% roll-off,

$$k^* = \arg \max_{k \geq 1} P_k, f_{dom} = f_{k^*}, A_{dom} = A_{k^*}, \quad (S10)$$

$$C = \frac{\sum_k f_k P_k}{\sum_k P_k}, S = \sqrt{\frac{\sum_k P_k (f_k - C)^2}{\sum_k P_k}}, F = \frac{\left(\prod_k P_k\right)^{1/K}}{\frac{1}{K} \sum_k P_k}$$

$$H_f = -\frac{1}{\ln 50} \sum_k p_k \ln p_k, p_k = \frac{P_k}{\sum_{\kappa} P_{\kappa}}, f_{85} = \min \left\{ f_k : \sum_{\kappa \leq k} P_{\kappa} \geq 0.85 \sum_{\kappa} P_{\kappa} \right\} \quad (S11)$$

Wavelet features. A four-level db2 decomposition with symmetric extension gives coefficient sets c , and their energy fractions are

$$\varepsilon_c = \sum_i c_i^2, EF_c = \frac{\varepsilon_c}{\sum_{c'} \varepsilon_{c'}}, c \in \{cA_4, cD_4, cD_3, cD_2, cD_1\} \quad (S12)$$

Eqs. (S8)–(S12) yield 24 frequency and wavelet features (8 amplitudes, 4 band shares, 7 spectral-shape features and 5 wavelet energy fractions). The dominant frequency takes a single value in the whole dataset and the roll-off frequency takes two values and is constant in the training cells of fold 0; their rank correlations are undefined in the affected folds and they are skipped by the finiteness condition of Section 2.3.3, which is why the feature-count ladder ends at 144 rather than 146.

Nonlinear features. Lagged autocorrelation ($\ell \in \{1, 5, 10\}$) and the three Hjorth parameters are

$$a = (V_0, \dots, V_{100-\ell}), b = (V_{\ell}, \dots, V_{100}), \rho_{\ell} = \frac{\sum_i (a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\sum_i (a_i - \bar{a})^2 \sum_i (b_i - \bar{b})^2}}, \quad (S13)$$

$$Act = \sigma^2(V), Mob(V) = \sqrt{\frac{\sigma^2(d^{(1)})}{\sigma^2(V)}}, Com = \frac{Mob(d^{(1)})}{Mob(V)}$$

Approximate and sample entropy use embedding dimension $m=2$ (and $m+1=3$), tolerance $r=0.2\sigma(V)$ and the Chebyshev distance; approximate entropy counts self-matches, whereas sample entropy counts only pairs $i < j$:

$$C_i^m(r) = \frac{\#\{j: \max_{0 \leq k < m} |V_{i+k} - V_{j+k}| \leq r\}}{N - m + 1}, \Phi^m(r) = \frac{1}{N - m + 1} \sum_i \ln C_i^m(r), \quad (S14)$$

$$ApEn = \Phi^2(r) - \Phi^3(r), SampEn = -\ln \frac{A}{B}$$

Permutation entropy uses order 3 and delay 1, and the Hurst exponent is the slope of a variogram regression over lags $\ell \in \{1, 2, 4, 8, 16\}$:

$$PE = -\frac{1}{\ln 6} \sum_{\pi \in \Pi_3} p(\pi) \ln p(\pi), \tau(\ell) = \sqrt{\frac{1}{101-\ell} \sum_i (V_{i+\ell} - V_i)^2}, \quad (S15)$$

$H = \text{OLS slope of } \ln \tau(\ell) \text{ on } \ln \ell$

Transient-boundary features. Fifteen voltage differences between specified samples, including the steps at current changes ($V_1 - V_0, V_{31} - V_{30}, V_{41} - V_{40}$) and the net changes within segments ($V_{30} - V_1, V_{40} - V_{31}, V_{100} - V_{41}$), describe the transitions between segments. Three response times are defined on the rest segment $S^{(3)}$, with linear interpolation between samples and $\alpha \in \{0.10, 0.50, 0.90\}$:

$$p_t = \text{sgn}(V_{100} - V_{41})(V_t - V_{41}), t = 41, \dots, 100, t_\alpha = \min\{t : p_t \geq \alpha |V_{100} - V_{41}|\} \quad (S16)$$

Supplementary Note S2. PatchTST encoder equations

Each patch p_m of Eq. (12) is embedded by a shared linear map and a learnable positional embedding e_m , initialised from $N(0, 0.02^2)$ and trained with the network:

$$z_m^{(0)} = W_p p_m + b_p + e_m, W_p \in R^{D \times P}, Z^{(0)} = [z_1^{(0)}, \dots, z_M^{(0)}]^T \in R^{M \times D} \quad (S17)$$

The M tokens enter one Transformer encoder layer whose multi-head self-attention ($h=4$ heads, $d_k=D/h=8$) acts between patches:

$$Q_i = Z W_i^Q, K_i = Z W_i^K, V_i = Z W_i^V, \\ head_i = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i, MHA(Z) = [head_1; \dots; head_h] W^O \quad (S18)$$

The layer uses post-normalisation residual connections and a GELU feed-forward sublayer of width $2D=64$ with dropout 0.2:

$$Z' = LN(Z^{(0)} + MHA(Z)), Z^{(1)} = LN(Z' + FFN(Z')), \\ FFN(u) = W_2 GELU(W_1 u + b_1) + b_2 \quad (S19)$$

Layer normalisation, mean pooling over tokens and a linear projection give the 32-dimensional waveform embedding:

$$E_v(\tilde{v}) = \text{Drop}\left(\text{ReLU}\left(W_o \cdot \frac{1}{M} \sum_{m=1}^M LN(z_m^{(1)}) + b_o\right)\right) \in R^{32} \quad (S20)$$

91 **Supplementary Note S3. Statistical test families**

92 Holm correction was applied within the following prespecified families at $\alpha =$
93 0.05: (1) the 14 encoder pairings versus MLP + CNN; (2) the five supplementary
94 waveform branches versus A2; (3) the confirmatory ablations R0 – A1, R0 – A2, R0
95 – A3 and A3 – A4; (4) the three patch configurations versus R0; (5) the three
96 adaptation ablations versus R0; (6) the six feature-count levels versus $K = 50$,
97 separately for RMSE and MAE; and (7) the reproduced reference model versus A2
98 and versus R0 (an extended family whose adjusted p values are conservative). A1 –
99 A2 was a single prespecified comparison without correction. Marginal effects,
100 difference-in-differences, stratified increments, residual complementarity and low-
101 SOH bias are descriptive. Confidence intervals are conditional on the fixed split and
102 the trained models and do not cover the variability of new data collection or
103 repartitioning.

104 **Supplementary Note S4. Re-implementation used for the generalisation,**
105 **robustness and deployment experiments**

106 Apart from the main NMC/graphite results, the experiments of Section 4.6 were
107 run with an independent re-implementation of the protocol written in Python (NumPy,
108 scikit-learn 1.9 and PyTorch 2.14 on CPU, one thread per run), starting from the
109 publisher’s processed pulse files [31]. Cleaning reproduced the record counts exactly:
110 17,130 records in 2,855 groups before cleaning, 192 records removed at the 42nd
111 RPT, and 16,938 records in 2,823 complete groups (SOH 72.34–100%) after cleaning.
112 The 143 waveform features were implemented from the definitions in Supplementary
113 Note S1; details not fixed by those definitions, such as the quantile levels of the global
114 family and the nine voltage differences not listed in Note S1, may differ from the
115 original code, and the response-time levels were set to 0.5, 0.632 and 0.9. Feature
116 selection was nonetheless similar: in fold 0 it selected 16 frequency and wavelet, 14

segment, 11 difference-dynamics, 6 global, 1 boundary and 1 nonlinear feature together with the DCIR, which ranked 13th; 44 features were selected in every fold and 57 in at least one. The split permutation used NumPy’s default generator with seed 42, which reproduces the fold sizes but not necessarily the fold membership of the original runs. Architectures, loss, optimiser, scheduler, early stopping and seeds follow Sections 3.2–3.4 of the main text, with positional embeddings initialised from $U(-0.02, 0.02)$.

Under the original five-fold protocol the re-implementation gives $R0$ 1.427 and $A2$ 1.474 pp (reported: 1.377 and 1.500 pp). The waveform increment keeps its sign and significance (-0.046 pp, -0.070 to -0.023 ; $p < 0.001$; 42 of 64 cells), and cells 12 and 3 are again the best- and worst-estimated cells. Table S6 lists all re-implementation results.

NMC/graphite dataset. The UConn-ILCC NMC dataset [31] records the same 100-s pulses at nine nominal SOCs; the three SOCs used for LFP (20%, 50% and 90%) were retained, giving six conditions per RPT, 8,016 records in 1,336 cell–RPT groups of 44 cells and SOH from 41.2% to 100%. Because NMC cells have a higher resistance, the voltage-step and range limits of Eq. (2) were relaxed to 0.3 and 0.8 V; no record was removed. The main-text results come from the original implementation (five folds $\times$ three seeds, $K = 50$, $\lambda = 0.1$, as for LFP): $R0$ 1.968 pp, $A2$ 2.132 pp, $R0 - A2 = -0.163$ pp (Holm-adjusted $p = 0.0004$, 39/44) and $A1$ 2.468 pp. The DCIR correlates strongly with SOH ($\rho \approx -0.82$) but entered none of the 50 selected features because it is redundant with selected waveform features, so it was added to the $A1$ input separately, as the definition of $A1$ requires; $R0$ and $A2$ were run unchanged.

Difference between the re-implementation and the original implementation. The re-implementation found no waveform increment on NMC ($R0$ 2.030 pp, $A2$ 2.026

142 pp, $R0 - A2 = +0.004$ pp, $p = 0.90$). Diagnostics trace the difference to the primary
 143 branch features. The re-implementation computes its features on the absolute voltage
 144 and, on NMC, selects absolute-level features such as the pre-pulse and final voltages
 145 and voltage quantiles (six in fold 0, almost none on LFP). Because the NMC open-
 146 circuit voltage is sloped and capacity fade shifts the actual SOC of each pulse, the pre-
 147 pulse voltage tracks SOH (mean absolute Spearman correlation 0.93, LFP 0.36) and
 148 predicts the actual SOC deviation with $R^2 = 0.964$ (LFP 0.489). Once the primary
 149 branch holds this information, the dynamic information of the waveform branch partly
 150 overlaps with it: within each pulse condition the selected features explain 99.4% of
 151 the variance of the first ten principal components of ΔV (LFP 18.3%). With the 28
 152 absolute-level features removed from the candidates, the re-implementation gives A2
 153 2.098 pp, R0 1.940 pp and $R0 - A2 = -0.158$ pp (-0.205 to -0.112 , $p < 0.001$, 36/44),
 154 consistent with the original implementation. Using all nine SOC_s (24,048 records)
 155 does not change the re-implementation result ($R0 - A2 = +0.017$ pp, $p = 0.67$), so the
 156 difference is not a matter of data volume.

157 *Noise not seen in training.* When Gaussian noise is added only to the test
 158 voltages of models trained on clean data, both neural models deteriorate strongly
 159 ($\sigma_{\text{noise}} = 0.5$ mV: R0 4.37, A2 4.60 pp), because difference- and spectrum-based
 160 features computed on the smooth, 0.1-mV-resolution resampled records are sensitive
 161 to noise absent from training. Models should therefore be trained on data from the
 162 target sensor, or with matching noise augmentation, as in the matched-noise
 163 experiment of Section 4.6.

Supplementary Algorithm S1

Algorithm S1. Training and inference of the dual-branch SOH estimation model (handcrafted primary branch + supplementary pulse-waveform branch).

Input: cleaned record set $D = \{(V, x, y, \text{cell}, \text{rpt}, \text{soc}, \text{pulse})\}$ with 16,938 records of 64 cells; $F = 5$ folds, split seed 42, initialisation seeds $\{0, 1, 2\}$; $K = 50$, redundancy threshold 0.98, $\lambda = 0.1$.

Output: record-level test predictions and cell-macro errors for every fold and seed.

Offline training

1: Remove cumulative cycle count, RPT index, coulomb-counted SOC and SOC deviation from the 150 candidates $\rightarrow$ 146 selectable features (Section 2.3.2)

2: Permute the 64 cells with seed 42 and divide them into five disjoint test folds (13/13/13/13/12 cells)

3: In each fold, draw validation cells from the non-test cells with the fold index as offset and stride 5 (train/validation/test = 40/11/13, 41/10/13, 41/10/13, 41/10/13, 42/10/12)

4: **for** outer fold $f = 0, \dots, 4$ **do**

5: Split D by cell into D_{tr} , D_{va} , D_{te} ; all RPTs and condition records of a cell stay together

6: Compute Spearman ρ_j of the 146 candidates with SOH on D_{tr} only $\triangleright$ Eq. (4)

7: Select K features greedily by $|\rho_j|$, skipping rank correlation ≥ 0.98 with selected ones; fill to K if needed $\triangleright$ Eq. (5)

8: Fit medians, feature means/s.d., per-position waveform means/s.d., cycle-count and label statistics on D_{tr} only

9: Transform D_{tr} , D_{va} , D_{te} with these statistics $\triangleright$ Eqs. (6)–(10)

10: **for** seed $s \in \{0, 1, 2\}$ **do**

11: Initialise E_h , E_v and g with seed s ; AdamW (lr 10^{-3} , weight decay 10^{-4}), batch 256,

12: ReduceLROnPlateau (patience 5, factor 0.5), gradient clipping 5, at most 100 epochs, early-stopping patience 15

13: **for** each epoch and mini-batch **do**

14: Enable gradients on τ ; forward pass gives $\hat{y}^* \triangleright$ Eq. (11)

15: Compute $\partial \hat{y}^* / \partial \tau$ by automatic differentiation (create_graph = True)

16: $\mathcal{L} \leftarrow \text{MSE}(\hat{y}^*, y^*) + \lambda \cdot \{[\partial \hat{y}^* / \partial \tau]_+^2\}$; back-propagate, clip, update $\triangleright$ Eq. (13)

17: After each epoch, compute the cell-macro RMSE on D_{va} ; keep the best weights; stop after 15 epochs without improvement

18: Load the best-epoch weights and predict D_{te}

Online inference (one pulse record)

19: Reference the 101-point voltage to its first sample and standardise with the fold statistics $\rightarrow \tilde{v} \triangleright$ Eqs. (6), (7)

20: Compute the K selected features, impute and standardise, append standardised $\tau \rightarrow \mathbf{h} \triangleright$ Eqs. (8), (9)

21: $\hat{y}^* \leftarrow g([E_v(\tilde{v}); E_h(\mathbf{h})])$; SOH $\leftarrow 100(\sigma_y \hat{y}^* + \mu_y) \triangleright$ Eqs. (11), (10)

Metrics and statistics

22: Average the six condition predictions of each cell–RPT, compute per-cell RMSE and the cell-macro mean $\triangleright$ Eq. (14)

23: Paired differences: 5,000 fold-stratified cell bootstrap intervals; 5,000 two-sided paired sign-flip tests; Holm correction within families

205 **Supplementary Tables**

206 **Supplementary Table S1.** Ageing protocols of the cycling-condition groups.

Group	CC–CV charge rate (C)	CC discharge rate (C)	DOD (%)	Cells (IDs)
1	2	1.5	60	6 (2–7)
2	2.42	1.5	60	6 (8–13)
3	2	2	60	6 (14–19)
4	2	1.5	80	6 (20–25)
5	2.83	1.5	60	6 (26–31)
6	2.42	2	60	6 (32–37)
7	2.42	1.5	80	6 (38–43)
8	3.25	1.5	60	6 (44–49)
9	2	1.5	100	6 (50–55)
10	2	2	80	6 (56–61)
11	2.83	1.5	80	4 (62–65)

207 From the cell test log in the publisher’s public repository [28]. Charging is CC–CV and discharging
 208 CC; rates refer to the nominal 1.2 Ah, so 2.42C, 2.83C and 3.25C correspond to 2.9 A, 3.4 A and 3.9 A.

209 **Supplementary Table S2.** Input information of the two branches.

Input	Dimension	Waveform branch	Primary branch	Source
First-sample-referenced voltage sequence	101	✓	—	Single pulse
Waveform-derived features (train-fold selected)	49	—	✓	Single pulse
Matched-condition DCIR (selected)	1	—	✓	Single pulse
Standardised cumulative cycle count τ	1	—	✓	Cycle record
Nominal SOC, pulse direction	—	—	Candidate, not selected	Test setting
Coulomb-counted SOC, SOC deviation	—	—	Excluded	Requires reference capacity
Other condition records, past capacities, cell identity	—	—	—	—

210 ✓, input enters the branch; —, not used or not applicable.

211 **Supplementary Table S3.** Ablation configurations and reference models.

Category	Configuration	Handcrafted primary branch	Supplementary waveform branch	τ	λ	Trainings
Main	R0	MLP, 50 features	PatchTST (9, 4)	✓	0.1	15
Encoder matrix	15 pairings	MLP / ResNet / FT-Transformer	CNN / MLP / TCN / PW-Trans. / PatchTST	✓	0.1	225 (incl. R0)
Branch ablation	A2	MLP	—	✓	0.1	15

Category	Configuration	Handcrafted primary branch	Supplementary waveform branch	τ	λ	Trainings
Branch ablation	A1	DCIR only (concatenated)	PatchTST (9, 4)	✓	0.1	15
Factor ablation	A3	MLP	PatchTST (9, 4)	✓	0	15
Factor ablation	A4	MLP	PatchTST (9, 4)	—	0	15
Patch parameters	P1, P2, P3	MLP	PatchTST (5, 2), (5, 4), (9, 2)	✓	0.1	45
Adaptation	D1, D2, D3	MLP	Flatten head, + RevIN, raw voltage	✓	0.1	45
Feature count	$K = 10, 20, 30, 75, 100, 144$	MLP, K features	PatchTST (9, 4)	✓	0.1	90 ($K = 50$ reuses R0)
Published model	Nowacki et al. [28], reproduced	—	Raw ΔV , 5×100 MLP	—	—	30 models

Neural configurations use 5 folds $\times$ 3 seeds (465 trainings in total, plus 30 reproduced models).

Supplementary Table S4. Stratified waveform increment (R0 – A2, SOH pp).

Stratum	Level	A2 RMSE	R0 RMSE	Δ RMSE	95% CI	Relative change	Improved cells
True SOH	<80%	2.404	1.991	−0.413	−0.460 to −0.363	−17.2%	62/64
True SOH	80–85%	1.742	1.601	−0.141	−0.200 to −0.082	−8.1%	47/64
True SOH	85–90%	1.974	1.861	−0.113	−0.151 to −0.074	−5.7%	50/64
True SOH	90–95%	1.150	1.123	−0.027	−0.046 to −0.009	−2.3%	36/64
True SOH	$\geq 95\%$	0.956	0.808	−0.148	−0.163 to −0.134	−15.5%	61/64
Nominal SOC	20%	1.546	1.452	−0.094	−0.116 to −0.072	−6.1%	55/64
Nominal SOC	50%	1.550	1.414	−0.136	−0.161 to −0.111	−8.8%	58/64
Nominal SOC	90%	1.579	1.474	−0.104	−0.127 to −0.080	−6.6%	56/64
Direction	Charge	1.546	1.434	−0.112	−0.136 to −0.088	−7.2%	55/64
Direction	Discharge	1.584	1.477	−0.107	−0.125 to −0.088	−6.7%	57/64
Condition	20% charge	1.533	1.431	−0.102	−0.128 to −0.076	−6.7%	47/64
Condition	20% discharge	1.556	1.471	−0.085	−0.109 to −0.061	−5.5%	49/64
Condition	50% charge	1.512	1.380	−0.132	−0.161 to −0.104	−8.7%	57/64
Condition	50% discharge	1.578	1.439	−0.139	−0.167 to −0.110	−8.8%	54/64
Condition	90% charge	1.567	1.458	−0.109	−0.141 to −0.078	−7.0%	52/64
Condition	90% discharge	1.584	1.483	−0.101	−0.124 to −0.077	−6.4%	54/64

True-SOH strata are formed at the cell–RPT level after averaging the six conditions, and each cell’s RMSE is computed within the stratum; SOC, direction and condition strata use single records. Predictions are averaged over the three seeds. Descriptive analysis without multiplicity correction.

Supplementary Table S5. Sensitivity to the number of handcrafted features (SOH pp).

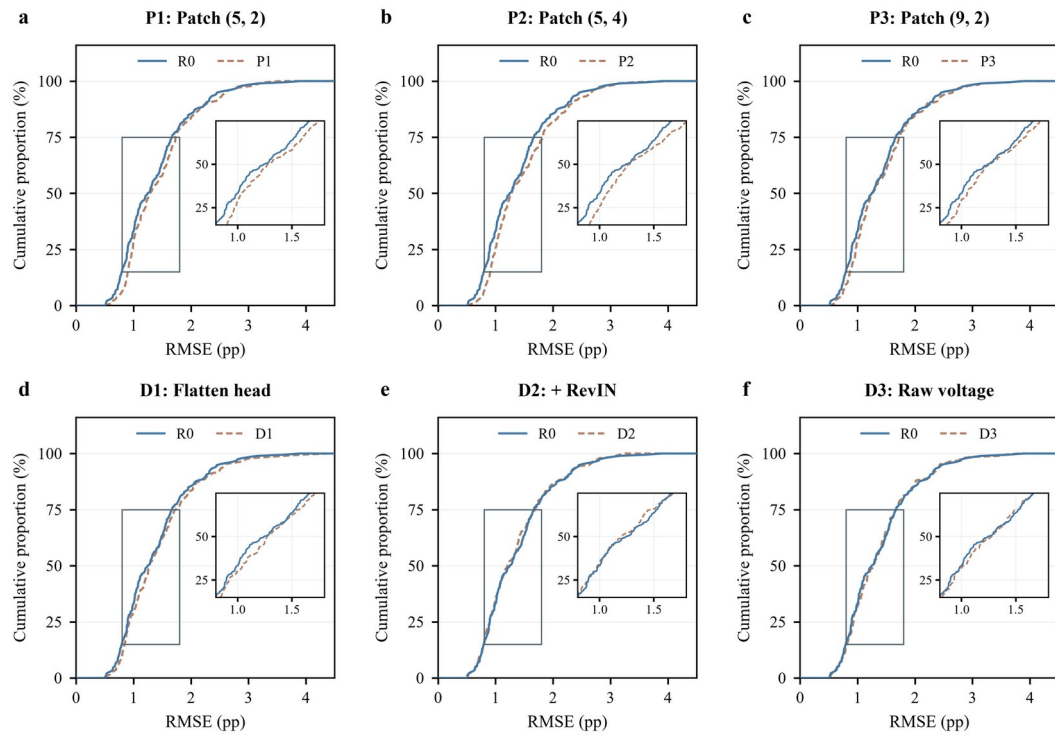
K	Primary -branch input	Parameters	RMSE	MAE	Δ RMSE	95% CI	Holm p	Improved / worse cells
10	11	16,481	1.516	1.249	+0.139	0.089–0.188	0.001	16/48
20	21	17,121	1.499	1.226	+0.122	0.077–0.169	0.001	14/50
30	31	17,761	1.468	1.210	+0.091	0.057–0.125	0.001	14/50
50 (R0)	51	19,041	1.377	1.141	—	—	—	—
75	76	20,641	1.414	1.165	+0.037	0.018–0.057	0.002	24/40
100	101	22,241	1.418	1.176	+0.041	0.021–0.062	0.002	15/49
144	145	25,057	1.376	1.128	−0.002	−0.019 to 0.015	0.838	32/32

K excludes the separately appended cycle count, so the primary branch input has $K + 1$ dimensions. Δ RMSE is the level minus $K = 50$; Holm correction covers the six RMSE comparisons. Improved (worse) cells have lower (higher) error than $K = 50$.

Supplementary Table S6. Results of the supplementary experiments (cell-macro RMSE, SOH pp).

Scenario	Data	R0	A2	A1	R0 – A2	95% CI	p	Improve d cells
Five-fold, reported (main text)	LFP	1.377	1.500	1.494	−0.123	−0.141 to −0.103	<0.00 1	58/64
Five-fold, re- implementation	LFP	1.427	1.474	—	−0.046	−0.070 to −0.023	<0.00 1	42/64
Leave-one-group-out	LFP	1.560	1.594	—	−0.035	−0.053 to −0.015	0.008	39/64
Cycle count +10% (test)	LFP	1.449	1.497	—	−0.048	−0.072 to −0.024	<0.00 1	43/64
Cycle count −10% (test)	LFP	1.477	1.518	—	−0.041	−0.063 to −0.018	0.002	39/64
Matched noise, $\sigma_{\text{noise}} =$ 1 mV	LFP	1.620	1.678	—	−0.058	−0.077 to −0.040	<0.00 1	50/64
Matched noise, $\sigma_{\text{noise}} =$ 2 mV	LFP	1.715	1.760	—	−0.045	−0.069 to −0.022	0.002	46/64
Test-only noise, σ_{noise} = 0.5 mV ^a	LFP	4.374	4.595	—	—	—	—	—
Five-fold, original implementation ^b	NM C	1.968	2.132	2.468	−0.163	to be added	0.000 4 ^c	39/44
Five-fold, re- implementation	NM C	2.030	2.026	2.317	+0.004	−0.042 to 0.047	0.90	21/44
Five-fold, re- implementation, primary branch without absolute-level features	NM C	1.940	2.098	—	−0.158	−0.205 to −0.112	<0.00 1	36/44

Paired comparisons use 5,000 fold- (or group-) stratified cell bootstraps and two-sided sign-flip tests, without multiplicity correction. ^a Models trained on clean data with noise added to the test voltages only. ^b Original implementation; DCIR added to the A1 input (Supplementary Note S4). ^c Holm-adjusted. Group-level leave-one-group-out errors are shown in Fig. 7a of the main text.



228
 229 **Fig. S1.** Cumulative error distributions of the waveform-branch design variants. (a–c) Patch
 230 configurations P1 (5, 2), P2 (5, 4) and P3 (9, 2) versus R0 (9, 4); (d–f) adaptation ablations
 231 D1 (flatten head), D2 (with RevIN) and D3 (raw voltage). Solid blue, R0; dashed brown,
 232 variant. Each curve contains the 192 cell $\times$ seed RMSEs (not 192 independent cells); insets
 233 magnify RMSE 0.8–1.8 pp and cumulative proportion 15–75%. Cell-level tests are given in
 234 Table 8.

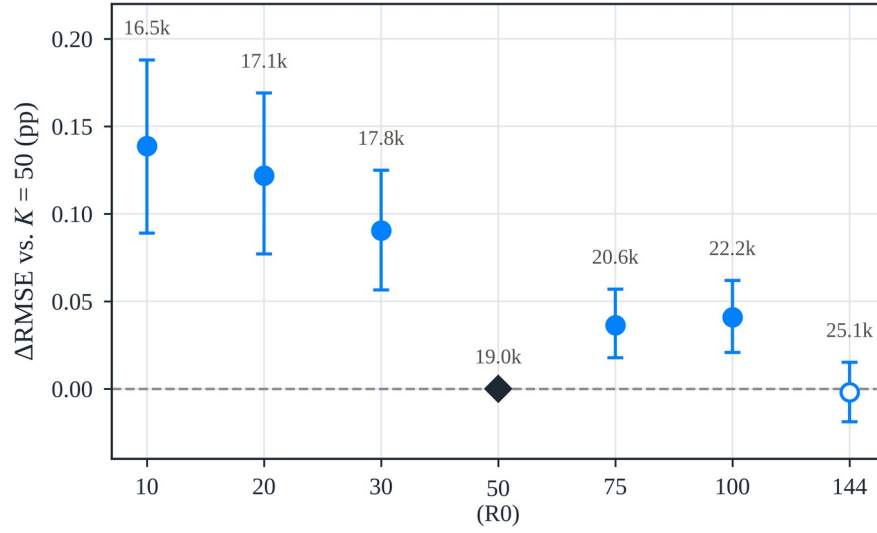


Fig. S2. Sensitivity to the number of handcrafted features K . Difference in cell-macro RMSE relative to $K = 50$ with 95% fold-stratified cell bootstrap intervals; labels give the total parameter count. Hollow markers denote intervals that include zero.

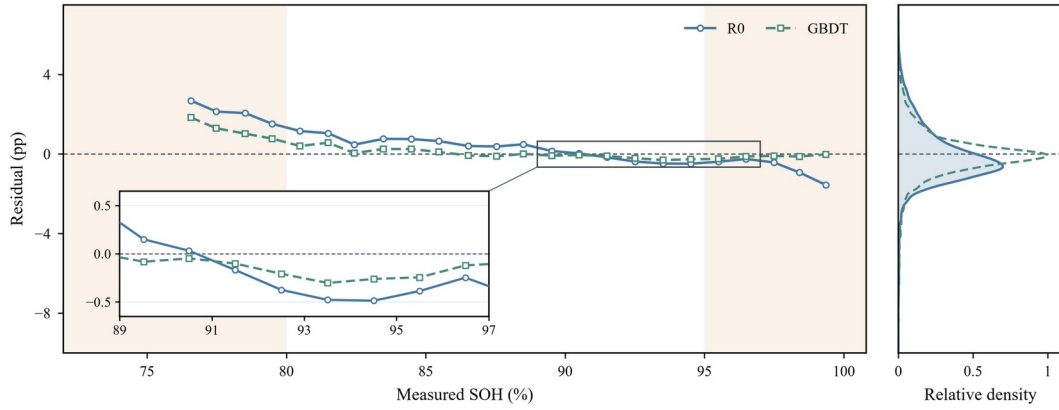


Fig. S3. SOH-dependent mean bias and overall residual distribution of R0 (blue, solid) and GBDT (teal, dashed). Residuals are the seed-averaged prediction minus the measured SOH of each record. Left: 1-pp bins of measured SOH with at least 20 records; shading marks SOH < 80% and $\geq 95\%$; the inset magnifies 89–97% SOH. Right: kernel density of all 16,938 residuals, normalised to the common maximum.

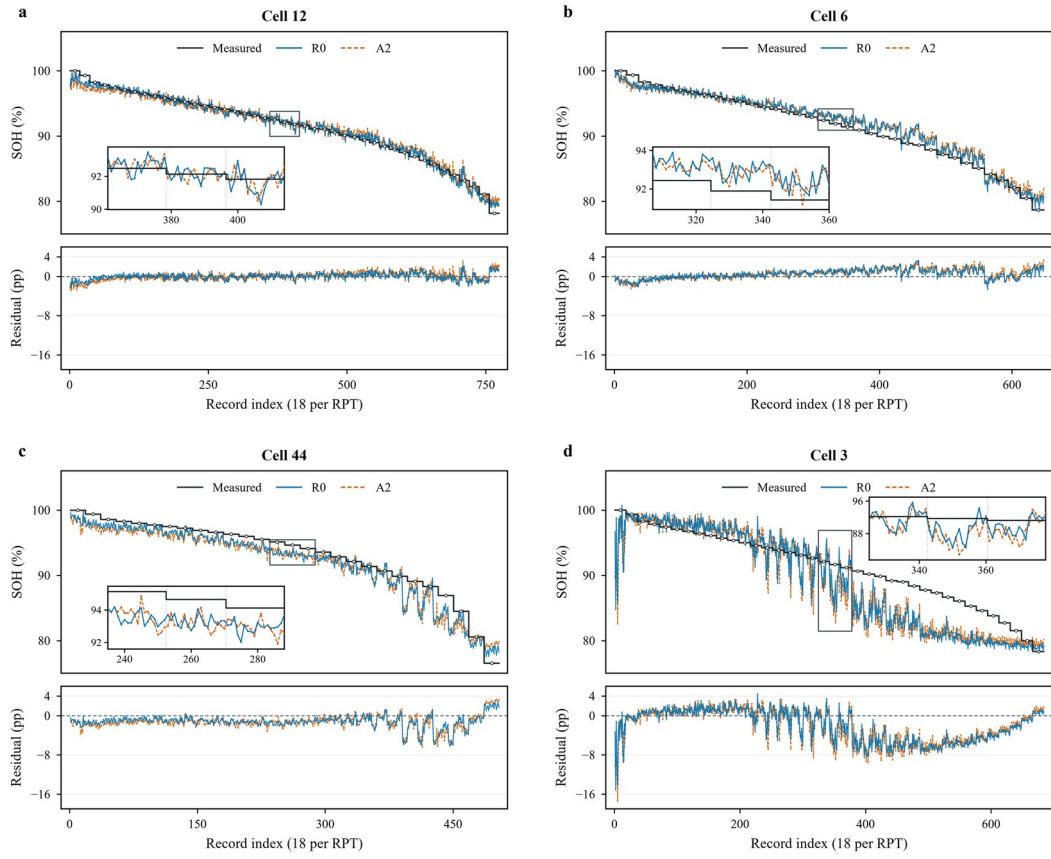
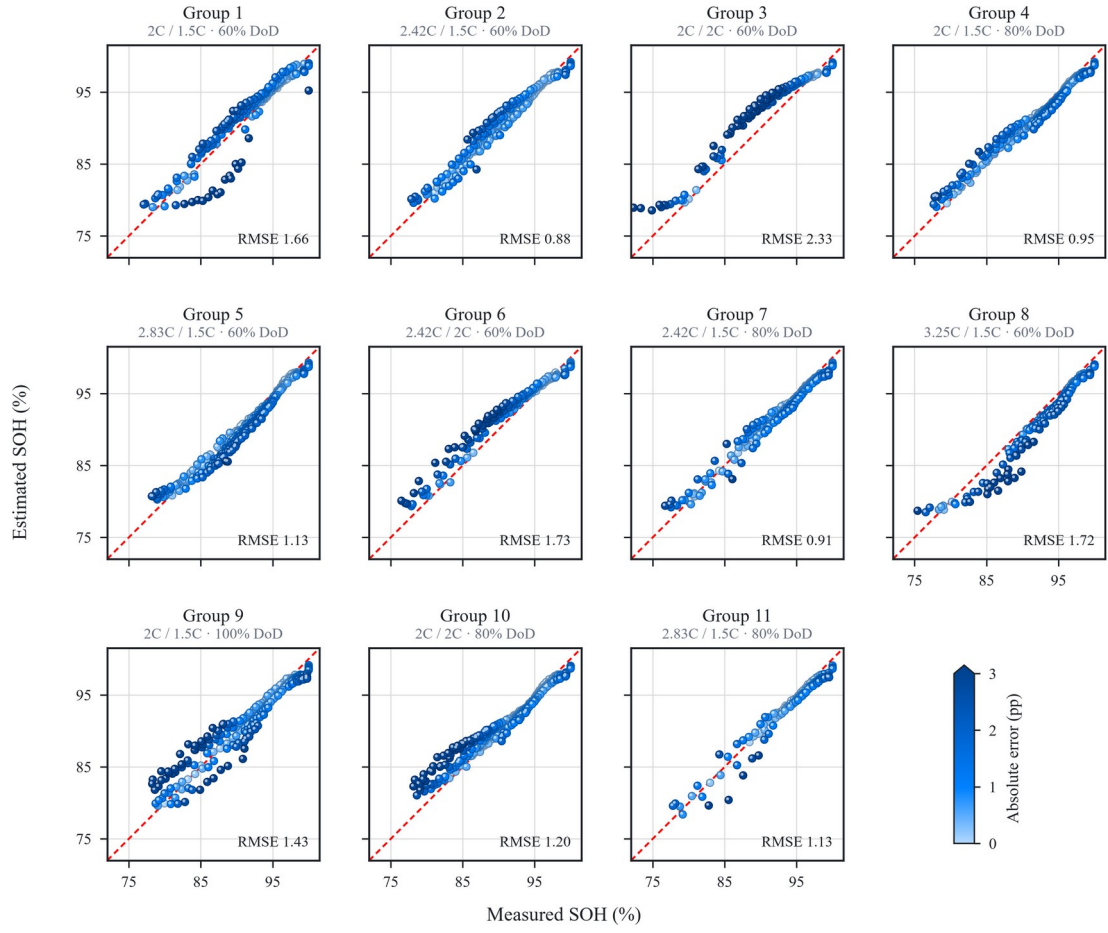


Fig. S4. Record-level predictions and residuals of representative cells 12, 6, 44 and 3 (ranks 1, 22, 43 and 64 by R0 error). Each RPT contributes 18 records (three SOCs $\times$ two directions $\times$ three seeds), so the horizontal axis is the record index rather than the cycle count; the grey staircase repeats the measured SOH of each RPT. Blue, R0; orange dashed, A2.



250
 251 **Fig. S5.** SOH estimated by R0 versus measured SOH for the 11 cycling-condition groups.
 252 Each point is one cell-RPT (predictions averaged over seeds and conditions, all from the
 253 cell's test fold); colour denotes absolute error (saturating at 3 pp) and the red dashed line is y
 254 $= x$. The number in each panel is the group's cell-macro RMSE following Eq. (14).